YOLO-V3 Based Real-time Drone Detection Algorithm

Hamid R. Alsanad, PhD
Amin Z sadik, PhD
University of Al-Anbar /Al-Anbar-Iraq
Monash University/Melbourne-Australia
hamidsend@yahoo.com,
amin.sadik1@monash.edu

Abstract:

Drones are currently being used in a wide range of useful tasks that are too dangerous or/and expensive to be performed by humans. However, this is increasingly developing security breaching issues due to the possibility of misuse of unmanned aircraft in illegal activities such as drug smuggling, terrorism etc. Thus,thedetection and tracking of dronesare becoming a crucial topic. Unfortunately, due to the drone's small size, its' detection methods are generally unreliable: high false alarm rate, low accuracy rate and low detection speed are well-known aspects of this detection.

The newemerging real-time algorithm based on the improved "You Only Look Once - version 3" (YOLO-V3) algorithm is proposed here for drone detection. This newly designed algorithm is consisting of three phases and has shown the potential to outperform the traditional detection approaches. The newly designed algorithm is trained and evaluated on the designed drone dataset. The evaluation results of our algorithm obtain 96% on average precision and 95.6% on accuracy.
Keywords: Real-time, Drone, Detection System, Detection Algorthim.

1. Introduction and Related Work

One of the most essential tasksin computer vision is the object detection that determines the presence or absence of definite features in an image [1][2]. When features are detected, this means that an object has been detected and that object can be classified as belonging to one of the pre-defined groups of categories, and then a bounding box around that object or the central point of the object is expected.In general, there are approximately three kinds of object detectors, namely:

Classic Detectors: Classic detectors operate on the principles of the sliding window, in which every time it applies a classifier over a pre-defined image grid. The most famous of them are the face detector by Viola and Jones [3]; CNN for digits recognition by LeCun et al. [4]; and the HOG method for pedestrian detection [5]. Later with the development of deep learning, they have been outperformed by two-stage detectors, described next.

Two-stage Detectors: More modern methods use region proposal methods to create a scattered set of candidate proposals that should include all objects while filtering out most negative locations in an image first and then run a classifier on these proposals to separate them into foreground classes/background.This two-stage detection is the controlling type nowadays. Region-based CNNs (R-CNN) have got bigger over the last years with many refinements [6]–[8] and many extensions [15–17].Faster R-CNN [8] is the fastest framework based on Region Proposal. It works on the entire image and is faster than its antecedents, the R-CNN [6], and Fast R-CNN [7]. The region proposals and classification are the two stages in the pipeline. Here the entire image is passed through CNN to generate a feature map. Then another CNN called RPN is applied to that feature map to produce object proposals and objectness scores. By applying resizable anchor boxes centered at the location of each pixel of the feature map, the proposals and scores are generated. Then, Region-of-Interest pooling is applied to those suggestions to bring the proposals of objects in the same region to the same size. At last, those proposals are threaded during the fully connected (FC) layer, having softmax and linear regression to classify the objects and produce bounding boxes.

One-stage Detectors: The third approach is the most recent to object detection, assumes a single detection phase or single-shot detection is close to human nature in object detection. The three dominant methods are SSD [12], YOLO [13]–[15], and RetinaNet[16]. Roughly in the same order, their effectiveness grows, SSD has less average precision (about 10-20%). The release 3 of YOLO has roughly the same precision as two-stage detectors, and RetinaNet until now is the most effective state-of-the-art object detector. However, YOLO-V3 is the fastest among them and still has acceptable precision.The performance of the basic YOLO model is more enhanced inYOLO-V3. For feature extraction three different scales are used.Darknet-53 is the fundamental architecture. The multi-layer scalingpermits predictions at three different scales. The layer that predicts the object's bounding box with the objectness score and class is the last layer. For objectness score it uses logisticregression. Instead of one softmax, independent logistic classifiersare used for each class. Overall YOLO-V3performs better than SSD and issimilar to Residual neural network(ResNet-152)[11] but istwice as fast [14].

The paper is arranged as follows. Section 2 presents abrief introduction of YOLO-V3. The proposed algorithm is described in Section 3. Section 4 presents the experimental results, datasets and analyses, and Section 5 provides the conclusion and future work.

2. Brief Introduction of YOLO-V3

The network of YOLO-V3 [14] is evolved fromthe networks of YOLO-V1[13] and YOLO-V2 [15]. The YOLOframework transforms the detection task into a regression task.Compared with the Faster R-CNN framework itdoes not need a proposal region, and it generatesthe coordinates of bounding box and probabilities of each class straight through regression.It is greatly rising the detection speed related to Faster R-CNN.

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The model of YOLO detection is shown in Figure1. Each imageis divided via the networkin the training set into S×S (for YOLO-V1,S = 7) grids. Each grid is responsible for detecting the target if the center of the target ground truth falls in that grid. Also, each grid expects B bounding boxes and confidence scores of those bounding boxes, in addition to C class conditional probabilities. Confidenceis defined as follows:

$$Confidence = P_r(Object) \times IoU^{truth}_{pred}, P_r(Object) \in \{0,1\} (1)$$

When the target is belongingto the grid,$P_r(Object) = 1$ and 0 otherwise. The intersection over unionIoU^{truth}_{pred}represents the coincidence between the ground truth box and the predictedbounding box. The confidence reflects whether the grid includes objects and the expected bounding box accuracy when it includes objects.When many bounding boxes detect the same object, the non-maximum suppression (NMS) method is used to choose the bestbounding box.

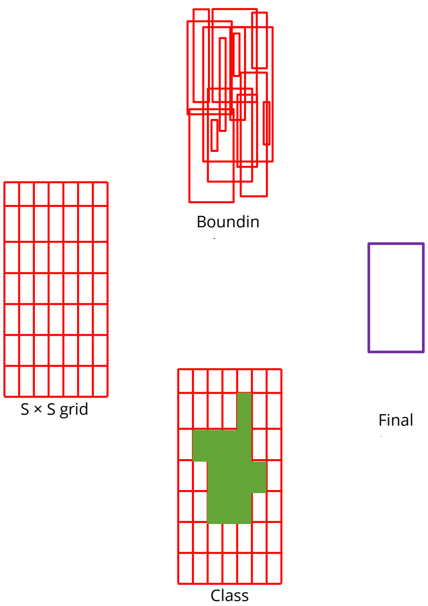

Figure 1. YOLO modelsdetection.

Although the first release (YOLO-V1) is much faster if we compare it to Faster RCNN, it has a significant detection error. For the purpose of overcoming this problem, release two (YOLO-V2) introducedthe anchor box idea and uses the k-means clustering method to generate appropriate prior bounding boxes. Consequently, the anchor boxes number required to obtain the similar IoU results decreases. YOLO-V2 develops the network structure and employs a convolution layer to substitute the FC layer in the YOLO-V1 output layer.YOLO-V2 also offers batch normalization, dimension clusters, a high-resolution classifier, fine-grained features, multi-scale training, direct location prediction, and other techniques that highly improve the accuracy of detection compared to YOLO-V1.

The third release (YOLO-V3) is the improved of YOLO-V2version. YOLO-V3 employs a multi-scale prediction technique to detect the final target, and the structure of its network is more complex than YOLO-V2.Because of YOLO-V3 predicts bounding boxes on various scales, this multi-scale prediction makes YOLO-V3 more efficient for detecting small objects than other releases of YOLO.

3. The Proposed Method

This section proposes three phasesof improvements to YOLO-V3framework to make it more suitable for drone detection.

The first phase presents an improvement to the structure of YOLO-V3 network. In YOLO-V3, the authorsintroduced the Darknet-53 network based on ResNet. Although Darknet53 with residual structure reduces the network training difficulty, reduces the number of parameters byusing a large number of (1×1) and (3 × 3) convolution kernels with step size of two instead of maximum pooling. The detection of single-class objectsby the Darknet53 network is somewhat too complex and excessive. Furthermore, many parameters will lead to decelerate the detection speed, increase the data volume demands, and the complexity of training.

To realize a real-time detection of the drones, the work hereindepends onDarknet53 to maintain accuracy and reduce the number of parameters as a starting point and proposes a CNN with a small number of parameters and relatively low computational complexity as a feature extraction network.

The CNN proposed in this paper is called Darknet49. In this neural network, a 1×1 convolution kernel is used in the transition module to further reduce the dimensionality. Since the use of a non-linear activation function for a low-dimensional convolutional layer will destroy image information to a certain extent, this paper uses a linear activation function in the first convolutional layer, as shown in Table 1.

Table 1 The network structure of Darknet49

Type	Filters	Size/Stride	Output
Convolutional	16	conv 3×3/2	208×208
Res.Block (1) ×4	16	conv 1×1/1	208×208
	32	conv 3×3/1	
Trans.Module	16	conv 1×1/1	104×104
	32	conv 3×3/2	
Res.Block (2) ×4	32	conv 1×1/1	104×104
	64	conv 3×3/1	
Trans.Module	32	conv 1×1/1	52×52
	64	conv 3×3/2	
Res.Block (3) ×4	64	conv 1×1/1	52×52
	128	conv 3×3/1	

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Type	Filters	Size/Stride	Output
Trans.Module	64	conv 1×1/1	26×26
	128	conv 3×3/2	
Res.Block (4) ×4	128	conv 1×1/1	26×26
	256	conv 3×3/1	
Trans.Module	128	conv 1×1/1	13×13
	256	conv 3×3/2	
Res.Block (5) ×4	128	conv 1×1/1	13×13
	256	conv 3×3/1	

The second phase is using densely connected convolutional networks (DenseNet) between layers. Due to convolution and down sampling the feature maps are reduced while training the neural network, and the feature information is lostthrough transmission. To make use of feature information more effective a (DenseNet) was proposed [17][18]. The DenseNetlinks each layer in feedforward manner to other layers; therefore,the input of the l layer is the concatenation of the outputs of the previous layers$x_0, x_1, \cdots\cdots, x_{l-1}$.

$$x_l = H_l\ (x_0, x_1, \ldots, x_{l-1}), (2)$$

Wherex_0 to x_{l-1}representsthe splice of thex_0 to x_{l-1}layers feature maps.While Hl isthecomposite function of activation function, convolutional layer, and batch normalization. In this work the dense connection to enhance the proposed Darknet49 network is used. The proposed Darknet49 network consists of 5 densely connected modules and 4 transition modules. There is a transition module between each densely connected module to reduce the feature map size. The output of the prior dense module,in the transition module, is separately pooled with a maximum step size of 2 and the output usinga2-step size convolution kernelis concatenated in series and used as the input of the next dense module. In this way, the connection between modules of the dense neural network is enhanced, reducing the loss of feature transfer between modules, and enhancing feature reuse.

The third phase presents an improvement to the multi-scale detection.In the standard YOLO-V3, the Feature Pyramid Networks(FPN) network [9] is introduced, which simultaneously uses the high resolution of low-level features and high semantic information of high-level features and fuse the features of different layers through up-sampling to detect objects on three different scale feature layers. In view of the fact that most of the dronesin images are small objects, this paper improves the *scale detection* module in YOLO-V3, expands the original three scale detections to four scale detections, and assigns more accurate anchor boxes to small targets in larger feature maps.Using the intersection ratio of rectangular boxes (IOU, expressed by RIOU) as the similarity, all targets of the remote sensing aircraft target training set are labeled using K-means clustering to obtain the size of the anchor box, and the distance function of K-means clustering is:

$$d\ (B, C)\ =\ 1 - R_{IOU}\ (B, C),(3)$$

Where B is the size of the rectangular frame, C is the center of the rectangular frame, $R_{IOU}\ (B, C)$ represents the overlap ratio of the two rectangular frames.This paper depends on the idea of DenseNet and up-samples the feature layers of the four scales to detect the corresponding multiples and then uses dense connection. The dense connection of scale detection layers further merges the features of different levels, enhancing the semantic information of each scale feature layer.Figure 2 shows the multi-scale detection module proposed in this workwhere 2 × indicates up-sampling with a step size of 2, 4 × indicates up-sampling with a step size of 4, and 8 × indicates up-sampling with a step size of 8.

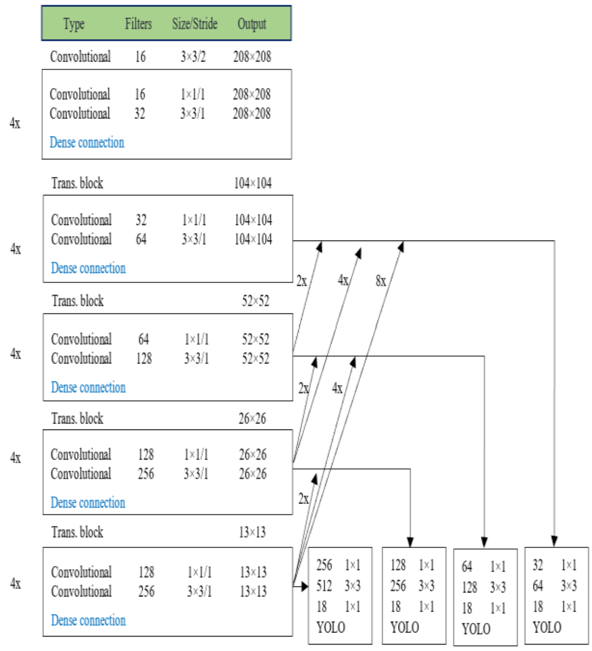

Figure 2. dense connection for multi-scale detection.

4. Experimental Results and Analyses

This section provides the details of the created datasetwith specifications along with the generated results,also this section details the comparison of theperformance ofboth models(YOLO-V3 and the proposed one).The algorithm in this paper is implemented on the computer configured with Intel(R) Core (TM) i7-6700HQ CPU @ 2.60 GHz CPU and NVIDIA GeForce GTX 965M GPU, with 16 GB memory and CUDA10.1 cuDNN9.1. The operating system of the computer is a 64-bit Windows 10. We adopted a MathWorks support example (Object Detection Using YOLO v2 Deep Learning) [19] to train our deep learning models. This example which had been pre trained on VOC 2007+2012 was selected to be the backbone of our CNN network. The results of this study verify the correctness and effectiveness of the proposed method in both accuracy and computational cost point of views.

4.1. Dataset Specification

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We have created our dataset by extracting5000 imagesof drones from a number of videos presented in global dataset recorded with the DJI Mavic Pro in various environments with and without presence of people[20]. This dataset is labeled using the Image Labeler application on MATLAB which is currently the most widely adopted labeler tool as shown in Figure 3. The Image Labeler application convertsthe labeled coordinates of the images directly to a mat file format contains the image's number, class(es) with boundary boxes.The labeled image is used as an input to train the proposed YOLOv3 based architecture.

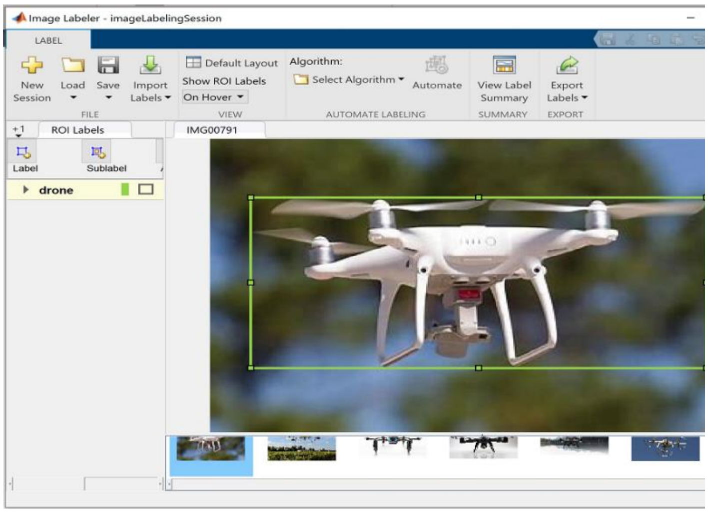

Figure 3.Sample labeling of drones using the Image Labeler application.

4.2. Precision, Recall, and F1 Score

The performance of the model is evaluated using recall rate, precision, F1-score and overlap ratio IOU.Precision is defined as the ratio of TP to the total positive predictions, expressed as:

$$P = \frac{TP}{FP + TP} \qquad (4)$$

The recall rate is:

$$R = \frac{TP}{FN + TP} \qquad (5)$$

The precision-recall (P-R, for short) curve, we can get it by representing the precision ratio as a vertical axis and the recall ratio as a horizontal axis. Another evaluation standard also used to test the model performance called F1-score, its definition is shown as:

$$F1 = \frac{2P \times R}{P + R} \qquad (6)$$

Where TP is a true positive example, FP is a false positive example, and FN is a false negative example.

In this paper, two indicators that evaluate the processing speed of the algorithm are used; one is the number of images processed per second by the algorithm, its unit is frame per second (f/s), and the other is the time required for the algorithm to process each image t = 1/N, the unit is ms.

4.3. Experiential Results

The dataset is trained for YOLO-V3 algorithm and the proposed algorithmseparately. For both algorithmstraining continueduntil the wanted average loss is reached and then all weights are tested after training. The weight that giveshigher mAP is selected for drone detection.Figure 4 shows the training loss curves for the two algorithms.

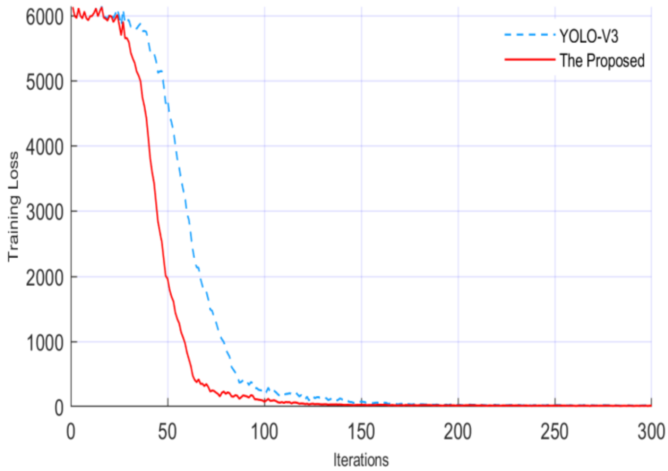

Figure4. Training loss curves for YOLO-V3 algorithm and the proposed algorithm.

Example ofthe resultsis shown inFigure 5 which illustrates how the system has successfully detected drones in different scenes and under different backgrounds.The YOLO-V3detection accuracy is 92.35% while the accuracy ofthe proposed based on YOLO-V3 is 95.60%. Moreover, YOLO-V3 took much longer trainingtime than the proposed one because darknet-53 is a deep networkwith 53 convolutional layers compared with darknet-49 of the proposed algorithm. The mAPis also calculated for both algorithmsby testing each weight. The weight with the highestmAP value is selected for testing. For YOLO-V3 the highest achieved mAP is 0.33after training on darknet-53 whileon the other hand, 0.36mAP is achieved after training on the proposedDarknet-49 for YOLO-V3. Table 2 summarizes all these improvements.

Table 2: Comparison of Performance for both Models

Model	(mAP)	F1-score	Accuracy%	FPS
YOLO-V3	0.33	86.43	92.35	35.55
The proposed	0.36	95.87	95.60	60.30

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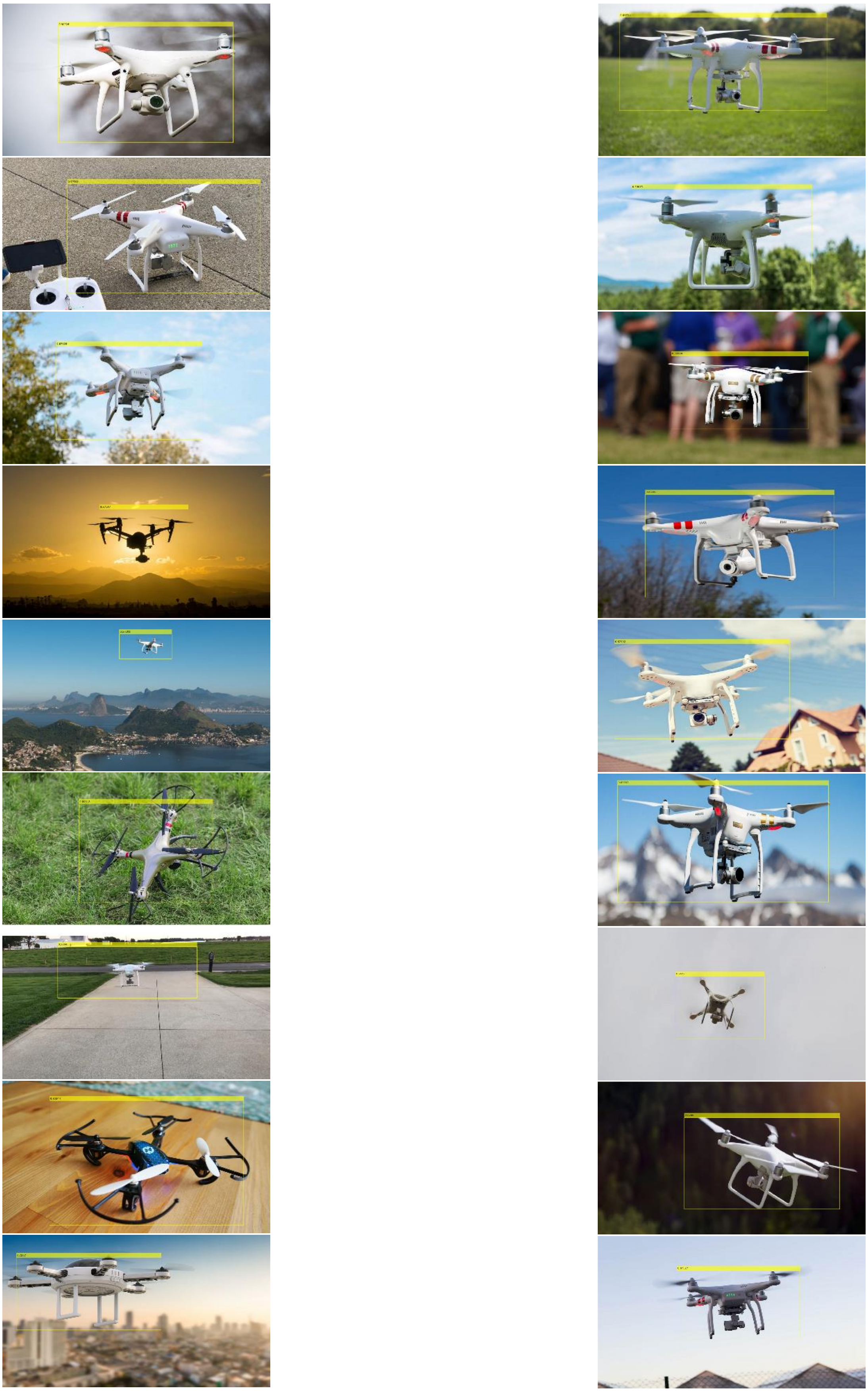

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Figure 5.Selected examples of drone detection results on the drone dataset using the proposed algorithm. Each output box is associated with the score, and the detected frame rate is displayed in the upper-left corner.

As shown in Figure 6, the proposed method has an Average Precision (AP) value of 96% on the testing set, which is higher than the standard YOLO-V3 method (92%).

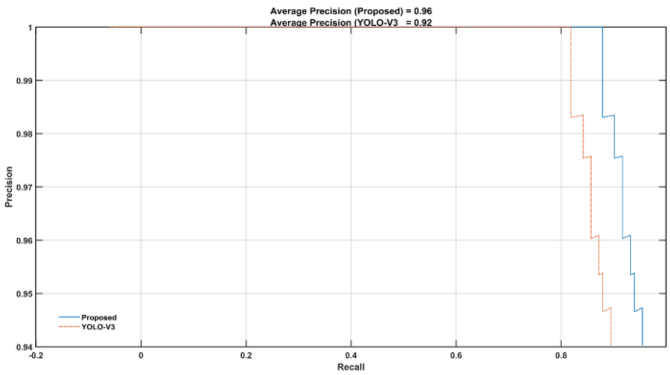

Figure6.Precision-recall curve performance of Proposed detection and YOLO-V3 Detection.

5. Conclusion

A real-time detection algorithm for dronedetection is proposed based on modification of YOLO-V3. The modified version includes improvements to the network structure and multi-scale detection of YOLO-V3. Moreover, designs and trains dataset of drone detection in images is implemented.

The experimental results illustrate that ourproposed algorithm has good robustness to drone detection. The accuracy and average precision have reached 95.60 and 96%, respectively.

This work has verified the feasibility and efficiency of YOLO-V3 for the detection of drone objects in the images.

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